Assignment 3: Data Pipelines

Big Data Engineering Spring 2025 Session

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Section 1: Introduction & Objectives

1.1 Broad Overview of the Project

The broad aim of this assignment is to build robust, production-ready ELT data pipelines that efficiently process and transform Airbnb and Census data for Sydney. Using Apache Airflow and dbt Cloud, the pipeline ingests raw data, applies layered transformations following a medallion architecture, and creates analytical data marts to enable insightful business analysis. This setup not only ensures scalable data management but also supports answering key market and policy questions about the short-term rental ecosystem.

1.2 Business Perspective

Airbnb's rapid expansion in Sydney has redefined lodging experiences by connecting millions of travelers with a variety of short-term accommodations beyond traditional hotels. Currently, Sydney hosts over 12,500 active Airbnb listings, generating an average annual revenue of approximately AU$70,000, with occupancy rates averaging 78%. This vibrant short-term rental market plays a significant role in the local economy, influencing tourism, real estate demand, and urban dynamics. Understanding the interplay between host demographics, revenue patterns, and housing market factors such as mortgage repayments is essential for balanced urban planning and policy-making. Analytical insights derived from integrating comprehensive Airbnb data with Census socioeconomic indicators empower stakeholders to make informed decisions that support sustainable growth while mitigating adverse community impacts.

1.3 Technical and Data Infrastructure Perspective

From a technical standpoint, working with large and varied datasets like Airbnb listings and Census information calls for a carefully architected and scalable data engineering solution. Using Apache Airflow allows automated and reliable orchestration of monthly data ingestion workflows, ensuring fresh data is continuously captured from various sources without manual intervention. dbt Cloud adds a layer of robust and modular data transformation using a medallion architecture, organizing data into raw (Bronze), cleansed (Silver), and analytics-optimized (Gold) layers. The integration of Slowly Changing Dimension (SCD) Type 2 logic within dbt ensures that historical changes are preserved accurately, enabling longitudinal analysis over time. Together, these tools create an efficient ELT pipeline that turns raw data into trustworthy and temporally consistent datasets, allowing for deeper insights like host mortgage coverage and neighborhood revenue performance, which is critical for guiding strategic decisions and urban policies in Sydney’s evolving Airbnb market.

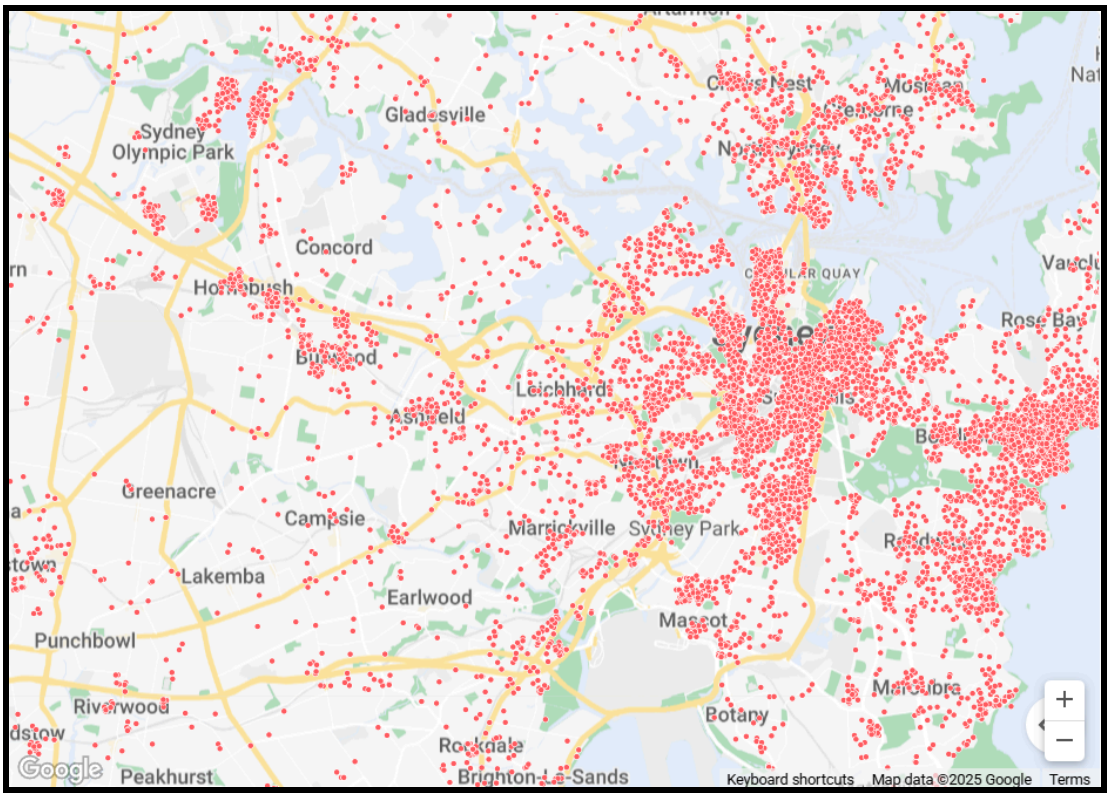


Figure 1: AirBnB Listing Spread Across Sydney (Source: airroi.com)

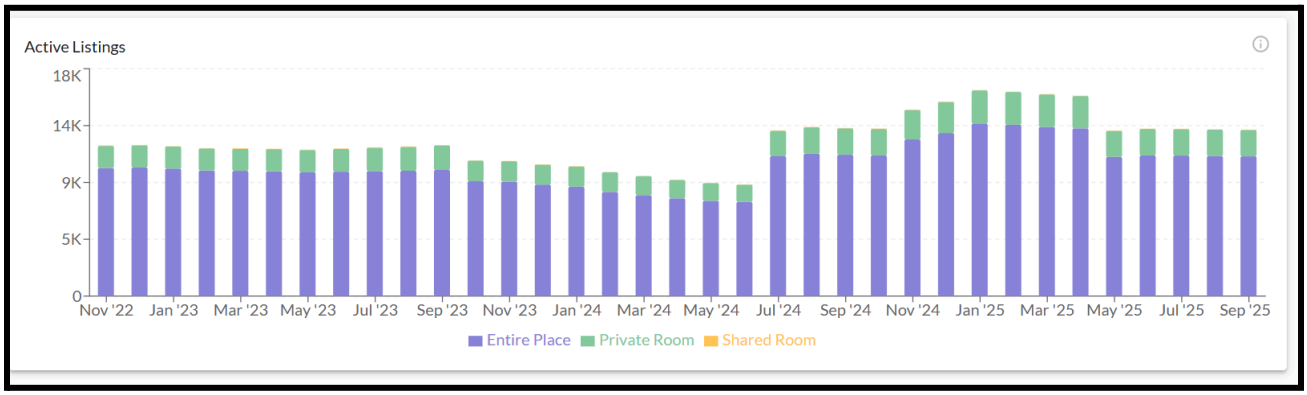


Figure 2: Monthly count of Active Listings Sydney (Source: airroi.com)

Section 2: Data and Infrastructure Overview

2.1 Dataset Overview

The analysis conducted in this project utilizes five pivotal dataset tables, each providing a crucial perspective for understanding Sydney's short-term rental market alongside demographic and socioeconomic contexts.

Airbnb Listings Data:
This dataset comprises detailed records of individual Airbnb listings in Sydney collected monthly from May 2020 through April 2021. Each record includes information such as listing_id, host_id, property attributes (e.g., property_type, room_type, accommodates), neighborhood identifiers (host_neighbourhood, listing_neighbourhood), availability metrics (has_availability, availability_30), pricing (price), and guest review scores across multiple dimensions (review_scores_rating, review_scores_cleanliness, etc.). This dataset captures the dynamic landscape of the Airbnb market, enabling granular insights into listing activity, pricing strategies, and host performance.

Australian Census Data (G01 & G02):
The tables G01 and G02, sourced from the 2016 Australian Bureau of Statistics Census, provide demographic and economic variables aggregated at the Local Government Area (LGA) level. The G01 dataset presents population distributions by age, gender, and cultural background, while G02 provides median statistics including age, mortgage repayments, personal and family income, rent, household size, and occupancy rates. These variables offer essential context linking neighborhood demographics and economic conditions to short-term rental market behavior.

LGA-Suburb Mapping Data:
This is a critical bridging dataset that maps Airbnb host neighborhoods (suburbs) to standardized LGA codes and names. This mapping enables accurate spatial joins between the Airbnb and Census data, facilitating aggregation and analysis at the LGA level. The dataset helps resolve discrepancies caused by variations in suburb naming and administrative boundaries, ensuring all analyses maintain geographic consistency.

LGA Codes:
The Local Government Area (LGA) codes are standardized five-digit numeric identifiers used to uniquely represent administrative regions across New South Wales, including Sydney. These codes are structured with a state or territory prefix followed by a unique LGA identifier, providing a precise and consistent way to reference geographic areas. They are essential for joining datasets consistently, especially linking Airbnb listing data at the suburb level with demographic and economic information aggregated at the LGA level.

2.2 ETL Infrastructure for the project:

The data pipeline is built using Apache Airflow, dbt Cloud, and PostgreSQL, all running on Google Cloud Platform (GCP).

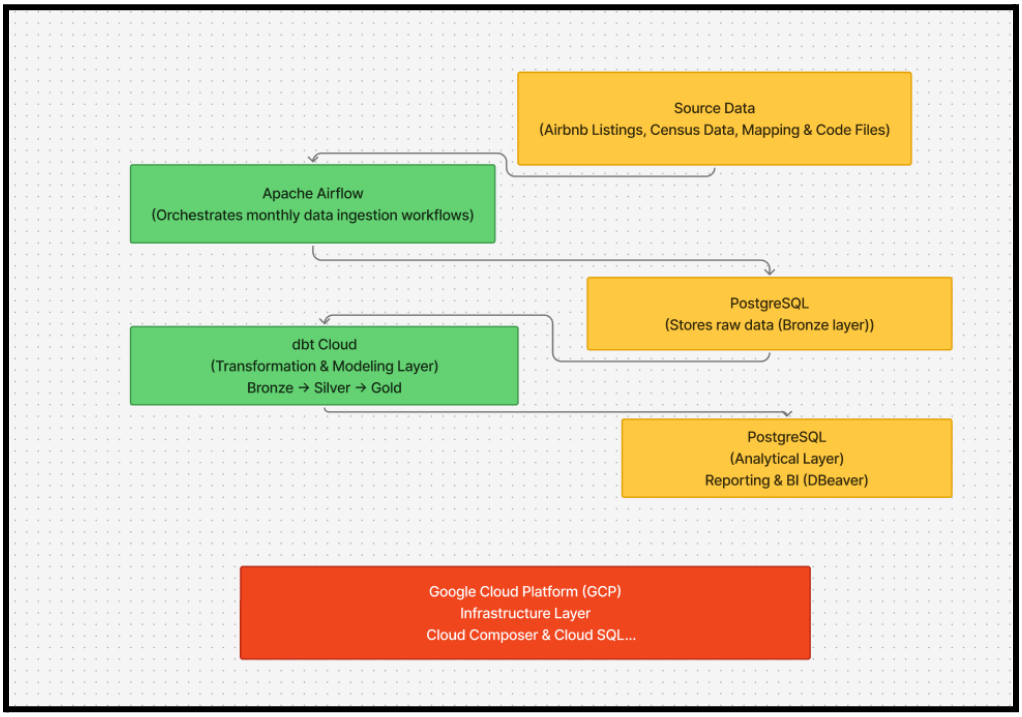
Apache Airflow:
It serves as the orchestration tool that manages end-to-end workflows. It uses Directed Acyclic Graphs (DAGs) to schedule and monitor monthly data ingestion tasks, moving raw Airbnb listings, Census data, and mapping files from source systems into the data warehouse.

Dbt Cloud:
It functions as the transformation and modeling layer. It enables modular SQL-based transformations that clean, validate, and enrich raw data as it progresses from the Bronze (raw) layer to the Silver (cleaned) and Gold (analytical) layers. dbt's built-in testing, documentation, and version control capabilities support collaboration and maintain consistency across data models.

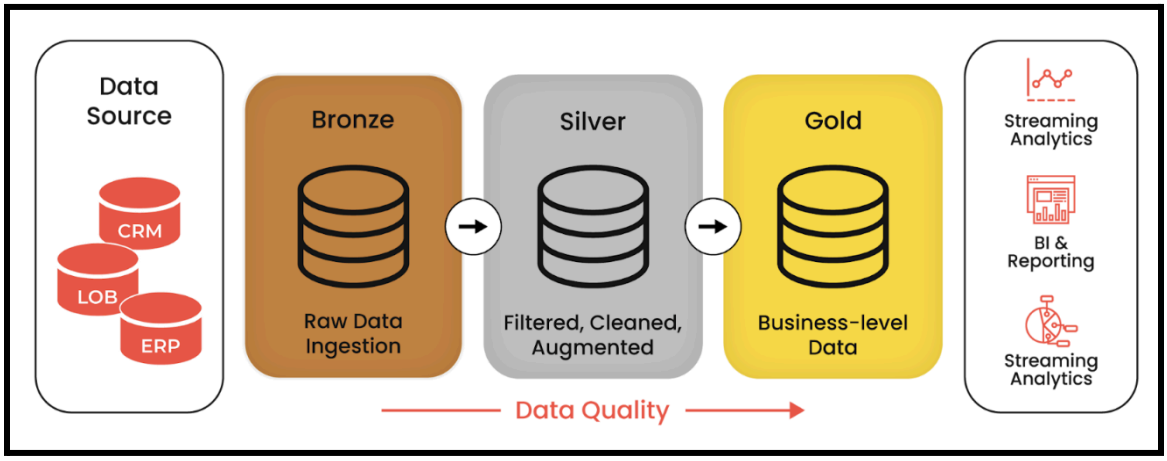
Dbeaver (PostgreSQL):
It acts as the central data warehouse, storing all datasets from raw to analytical. It supports efficient querying and integrates seamlessly with business intelligence tools for analysis and reporting.

The entire infrastructure operates on Google Cloud Platform (GCP), which provides managed services such as Cloud Composer (for Airflow orchestration) and Cloud SQL (for PostgreSQL hosting). This cloud-based setup ensures scalability, high availability, and security, which are essential for a production-grade ELT pipeline.

Figure 3: Infrastructure Flowchart(Tool: figma.com)



2.3 Medallion Architecture: Bronze, Silver & Gold



The Medallion Architecture is a widely used data engineering framework that organizes data pipelines into three progressive layers: Bronze, Silver, and Gold. This structure ensures data quality, scalability, and transparency throughout the transformation process.

Bronze Layer: Raw Data

The Bronze layer acts as the landing zone where all raw data from source systems first enters the pipeline. In this project, raw Airbnb listings, Census demographics, and LGA mapping and code files are ingested here from CSV sources using Apache Airflow workflows.

Silver Layer: Cleaned and Standardized Data

After ingestion, data moves into the Silver layer, where dbt Cloud transformations refine and standardize it. Here, data undergoes cleaning processes such as deduplication, validation, and standardization, along with joins across related datasets (for example, linking Airbnb listings to LGAs through suburb mapping).

Gold Layer: Business Analysis Ready Data

The Gold layer contains curated datasets optimized for analytics and reporting.
At this stage, data is aggregated into fact and dimension tables, often modeled in a star schema to represent key metrics such as total revenue, occupancy rates, and host demographics across neighborhoods or LGAs. Datasets in this layer are stable, reliable, and structured for business intelligence tools and dashboards, enabling stakeholders to generate insights and make informed decisions.

Snapshots were created to capture historical changes over time, enabling SCD Type 2 tracking of evolving host and neighborhood details. Dimension and fact tables were then built in the Gold

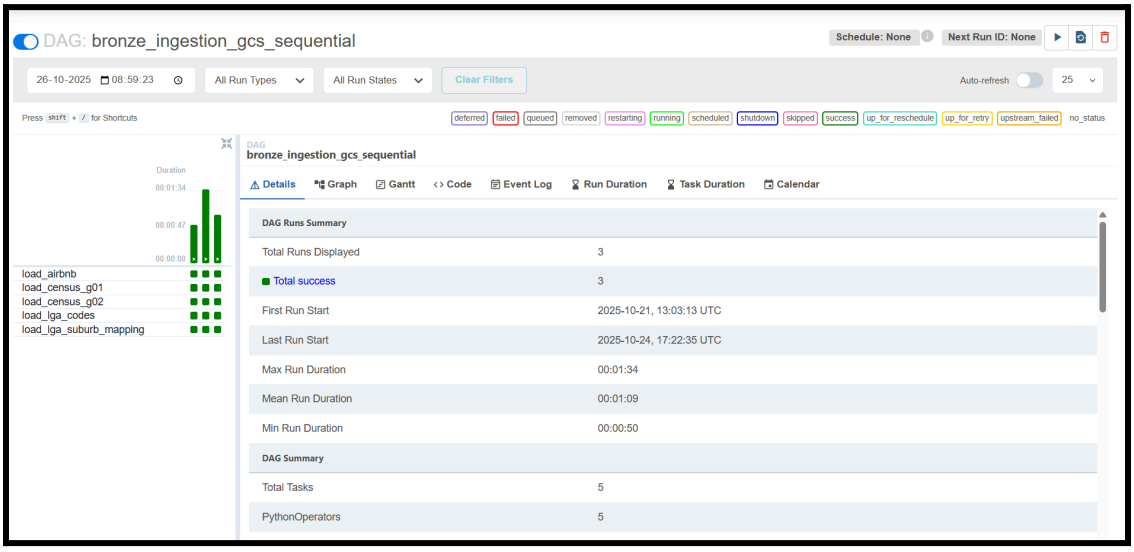
layer to store descriptive attributes and aggregated business metrics, optimizing the data for analysis and reporting.

Section 3: Data Ingestion with Airflow (Part 1 & 3)

The data ingestion process was designed to reliably bring raw Airbnb listings, Census data, and LGA mapping files into a PostgreSQL data warehouse, forming the Bronze layer of the ELT pipeline. This was achieved using Apache Airflow orchestrated on Google Cloud Platform (GCP) through Cloud Composer, with storage and initial staging handled via GCS buckets.

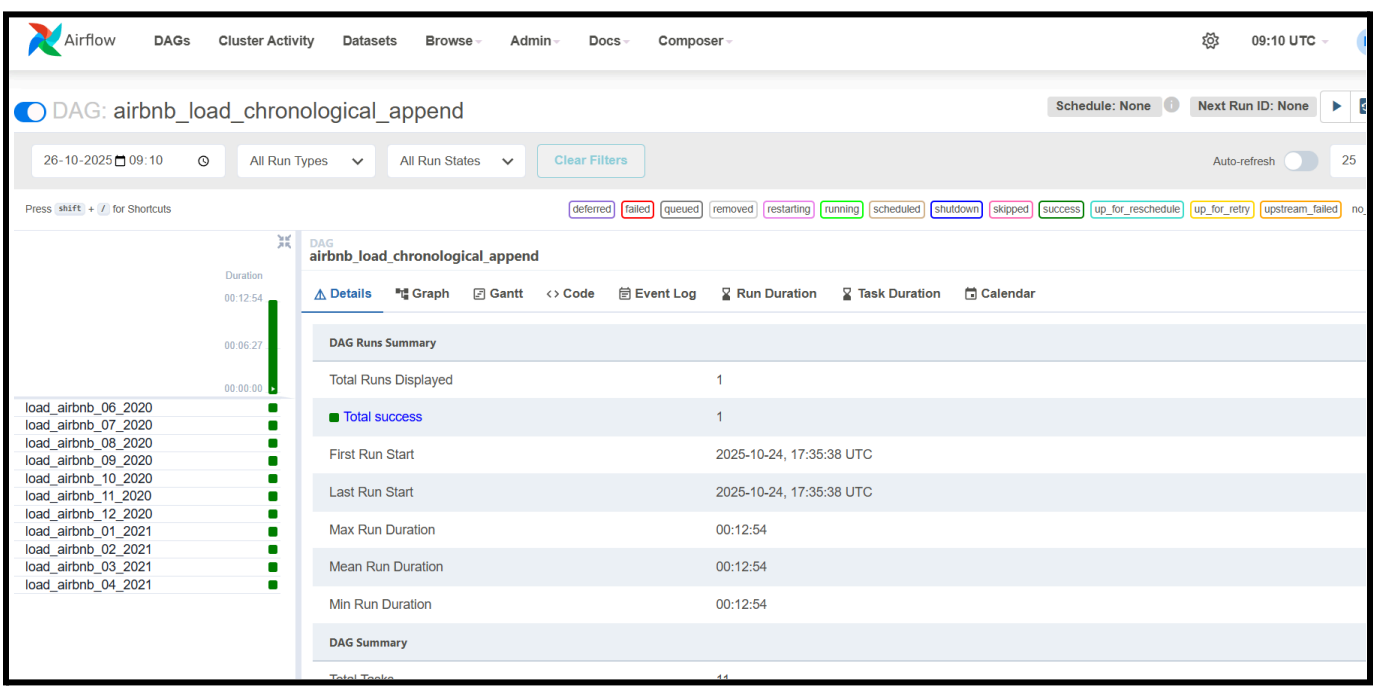
3.1 Part 1: Initial Data Loading:

1. The initial step of the data ingestion process involved uploading the May 2020 Airbnb listings (05_2020.csv), Census datasets (G01 and G02), and LGA mapping files to a dedicated Airflow storage bucket on GCP, which served as the landing zone for all raw data.
2. A Bronze schema was then created in PostgreSQL using DBeaver, with raw tables corresponding to each dataset to store the CSV data exactly as received, ensuring an immutable source of truth for traceability and auditing.
3. To orchestrate the ingestion, an Airflow DAG (bronze_ingestion_gcs_sequential) was built with no fixed schedule interval (schedule_interval=None) to allow manual execution. The DAG sequentially loaded datasets in the order of Airbnb data → Census G01 → Census G02 → LGA codes → LGA-suburb mapping.
4. Using PythonOperators, the DAG downloaded CSV files from GCS via GCSHook, cleaned the data by removing unnamed columns, standardizing column names, parsing date fields, converting boolean columns, and handling nulls, and then loaded the cleaned data into PostgreSQL, truncating tables initially to ensure a fresh start.
5. During this phase, only the first month of Airbnb data (May 2020) was loaded, allowing validation of the workflow, testing of data cleaning logic, and confirmation that the DAG could reliably process and store raw datasets end-to-end.



3.2 Part 3: Loading Remaining Airbnb Data:

1. After the initial Bronze ingestion, the Silver and Gold layers were designed and created using dbt Cloud to transform raw data into cleaned, standardized, and analytical datasets.
2. The Silver layer handled deduplication, validation, standardization, and table joins, producing reliable data with consistent schema and naming conventions. The Gold layer then aggregated metrics into fact and dimension tables, optimized for reporting and business intelligence.
3. Once these layers were in place, the DAG was extended to sequentially ingest the remaining Airbnb monthly datasets from June 2020 to April 2021. This DAG (airbnb_load_chronological_append) appended each month's data to the Bronze table, preserving chronological order to maintain historical integrity.
4. Each dataset was downloaded and cleaned using the same logic as the initial load and appended to PostgreSQL tables without truncation, ensuring all historical records remained intact.
5. Tasks were executed sequentially with manual triggering for controlled processing. To integrate the updated data into the pipeline, dbt jobs were created and manually triggered after each month's ingestion, transforming the Bronze updates into the Silver layer and propagating them to the Gold layer, maintaining a consistent, standardized, and analytics-ready dataset.



Section 4: Data Warehouse Design with dbt (Part 4)

A dedicated folder named models was created in dbt to organize all SQL transformations and data models. This folder contained separate subfolders for each layer of the pipeline Bronze, Silver, and later Gold. Each dataset was represented as a dbt model, with SQL scripts defining how raw data is ingested, cleaned, and transformed. dbt's materialized='table' configuration was used for Bronze and Silver models, ensuring that each model created a physical table in PostgreSQL, making the data directly queryable and viewable via DBeaver.

4.1 Bronze Layer: Raw Data Tables:

The Bronze layer highlighted the initial raw datasets in PostgreSQL without complex transformations. Models were created for each source:

1. Airbnb listings: bronze.airbnb_raw
2. Census data: bronze.census_g01, bronze.census_g02
3. LGA mapping files: bronze.lga_codes, bronze.lga_suburb_mapping

Each model was materialized using dbt's {{ config(materialized='table') }} to create persistent tables in the warehouse that are directly queryable. At this stage, the focus is on:

1. Organizing raw datasets consistently for further processing.
2. Keeping tables viewable for verification.
3. Maintaining a flow from the initial Airflow ingestion.

No deduplication, enrichment, or heavy cleaning was applied in dbt at this layer; the raw data had already undergone basic cleaning in the Airflow DAG (column standardization, date parsing, boolean conversion, and null handling).

4.2 Silver Layer: Cleaned and Standardized Tables

The Silver layer focused on transforming and standardizing the raw Bronze datasets in PostgreSQL. Models were created for each dataset:

1. Airbnb listings: silver.airbnb_raw
2. Census data: silver.census_g01, silver.census_g02
3. LGA mapping files: silver.lga_codes, silver.lga_suburb_mapping

Each model was materialized using dbt's {{ config(materialized='table') }} to create persistent tables in the warehouse that are directly queryable. At this stage, the focus was on:

1. Cleaning and standardizing data for consistency (trimming and lowercasing text fields, replacing nulls and invalid values, casting dates, numerics, and booleans).

2. Deduplicating and validating key fields to prepare dimensions like hosts, suburbs, and LGAs.
3. Ensuring tables are viewable in PostgreSQL for verification and downstream processing.
4. Establishing a high-quality, reliable dataset to serve as the foundation for Gold-layer analytical tables, while preserving historical integrity for slowly changing dimensions (SCD Type 2).

	listing_id	scrape_id	scraped_date	host_id	host_name	host_since	host_is_superhost	host_neighbourhood	listing_neighbourhood	property_type
1	11,156	20,200,000,000,000	2020-11-05	40,855	colleen	2009-09-23	[ ]	potts point	sydney	apartment
2	12,351	20,200,000,000,000	2020-10-05	17,061	stuart	2009-05-14	[ ]	pyrmont	sydney	townhouse
3	14,250	20,200,000,000,000	2020-11-05	55,948	heidi	2009-11-20	[v]	balgowlah	northern beaches	house
4	15,253	20,200,000,000,000	2020-11-05	59,850	morag	2009-12-03	[ ]	darlinghurst	sydney	apartment
5	44,545	20,200,000,000,000	2020-12-05	112,237	atari	2010-04-22	[v]	darlinghurst	sydney	loft
6	45,440	20,200,000,000,000	2020-10-05	201,611	deborah	2010-08-14	[v]	bondi beach	waverley	house
7	58,506	20,200,000,000,000	2020-11-05	279,955	john	2010-11-06	[ ]	mosman	mosman	house
8	61,721	20,200,000,000,000	2020-11-05	299,170	eilish	2010-11-25	[ ]	bondi beach	waverley	apartment
9	65,126	20,200,000,000,000	2020-10-05	318,390	nicolette	2010-12-14	[ ]	bondi beach	waverley	apartment
10	66,009	20,200,000,000,000	2020-10-05	322,887	belinda	2010-12-19	[ ]	paddington	woollahra	apartment
11	66,158	20,200,000,000,000	2020-12-05	323,771	maxine	2010-12-20	[ ]	elizabeth bay	sydney	apartment
12	66,170	20,200,000,000,000	2020-11-05	16,474	anand	2009-05-10	[ ]	ultimo	sydney	apartment
13	67,112	20,200,000,000,000	2020-11-05	160,705	liz	2010-07-07	[ ]	marrickville	inner west	house
14	68,999	20,200,000,000,000	2020-12-05	333,581	bryan	2011-01-03	[v]	sydney	hornsby	house
15	69,121	20,200,000,000,000	2020-12-05	345,292	pamela	2011-01-12	[ ]	unknown	northern beaches	apartment
16	69,956	20,200,000,000,000	2020-12-05	352,614	ben	2011-01-18	[v]	bardwell valley	bayside	guest suite
17	73,628	20,200,000,000,000	2020-10-05	385,108	dan	2011-02-10	[ ]	bondi beach	waverley	apartment
18	73,639	20,200,000,000,000	2020-10-05	17,061	stuart	2009-05-14	[ ]	pyrmont	sydney	townhouse

Figure 4: Post Silver Layer Cleaned Data: Command: ***select*** *** ***from*** *dbt_mmenon_silver.airbnb_raw ar*

census_g01 1

select * from dbt_mmenon_silver.census_g01 cg

	lga_code_2016	tot_p_p	tot_p_m	tot_p_f	age_0_4_yr_m	age_0_4_yr_f	age_0_4_yr_p	age_5_14_yr_m	age_5_14_yr_f	age_5_14_yr_p	age_15_1
1	LGA10050	51,076	24,662	26,411	1,689	1,594	3,286	3,208	3,117	6,328	
2	LGA10130	29,449	14,227	15,220	844	825	1,669	1,833	1,833	3,667	
3	LGA10250	41,790	20,127	21,658	1,074	997	2,072	2,565	2,294	4,856	
4	LGA10300	2,287	1,177	1,115	80	71	143	172	162	336	
5	LGA10470	41,300	20,695	20,605	1,339	1,209	2,546	2,883	2,691	5,571	
6	LGA10550	33,253	16,252	17,001	757	718	1,468	1,944	1,844	3,789	
7	LGA10600	12,668	6,123	6,548	303	304	605	791	801	1,594	
8	LGA10650	8,462	4,224	4,236	249	219	471	495	482	972	
9	LGA10750	336,962	167,574	169,390	13,683	13,245	26,928	25,722	24,010	49,739	
10	LGA10800	5,955	2,971	2,988	167	184	351	435	441	876	
11	LGA10850	7,257	3,613	3,647	235	212	451	540	520	1,058	
12	LGA10900	76,904	37,119	39,782	2,167	2,067	4,232	5,056	4,782	9,837	
13	LGA10950	2,692	1,355	1,334	91	115	207	182	161	344	
14	LGA11100	46,654	23,229	23,420	1,495	1,405	2,895	2,586	2,434	5,024	
15	LGA11150	2,634	1,301	1,329	109	109	222	202	195	395	
16	LGA11200	1,651	849	802	69	53	124	131	125	255	
17	LGA11250	17,708	8,615	9,091	496	478	974	985	981	1,972	
18	LGA11300	36,809	17,785	19,028	850	818	1,670	1,565	1,567	3,129	

Figure 5: Post Silver Layer Cleaned Data: Command: ***select*** *** ***from*** *dbt_mmenon_silver.census_g01 cg*

census_g02 1

select * from dbt_mmenon_silver.census_g02 cg

	lga_code_2016	median_age_persons	median_mortgage_repay_monthly	median_tot_prsnl_inc_weekly	median_rent_weekly	median_tot_fam_inc_weekly	average_num_psns_per_bedroom
1	LGA10050	39	1,421	642	231	1,532	
2	LGA10130	36	1,393	561	250	1,465	
3	LGA10250	48	1,733	601	340	1,426	
4	LGA10300	41	950	624	150	1,438	
5	LGA10470	37	1,670	646	280	1,632	
6	LGA10550	51	1,343	538	250	1,242	
7	LGA10600	49	1,400	515	280	1,218	
8	LGA10650	49	1,083	524	180	1,251	
9	LGA10750	33	2,150	672	380	1,817	
10	LGA10800	43	1,000	580	150	1,360	
11	LGA10850	42	1,500	620	220	1,581	
12	LGA10900	44	1,842	688	350	1,866	
13	LGA10950	40	1,083	634	160	1,561	
14	LGA11100	35	2,400	720	460	1,921	
15	LGA11150	36	991	719	140	1,562	
16	LGA11200	34	433	439	90	923	
17	LGA11250	45	953	527	190	1,279	
18	LGA11300	33	2,200	552	500	1,725	

Figure 6: Post Silver Layer Cleaned Data: Command: ***select*** *** ***from*** *dbt_mmenon_silver.census_g02 cg*

lga_codes 1

select * from dbt_mmenon_silver.lga_codes lc

	lga_code	lga_name
1	10050	Albury
2	10130	Armidale Regional
3	10250	Ballina
4	10300	Balranald
5	10470	Bathurst Regional
6	10550	Bega Valley
7	10600	Bellingen
8	10650	Berrigan
9	10750	Blacktown
10	10800	Bland
11	10850	Blayney
12	10900	Blue Mountains
13	10950	Bogan
14	11100	Bayside
15	11150	Bourke
16	11200	Brewarrina
17	11250	Broken Hill

Figure 7: Post Silver Layer Cleaned Data: Command: ***select*** *** ***from*** *dbt_mmenon_silver.lga_codes lc*

lga_suburb_mapping 1

select * from dbt_mmenon_silver.lga_suburb_mapping lsm

	suburb_name	lga_name
1	AARONS PASS	MID-WESTERN REGIONAL
2	ABBOTSBURY	FAIRFIELD
3	ABBOTSFORD	CANADA BAY
4	ABERCROMBIE	BATHURST REGIONAL
5	ABERCROMBIE RIVER	BATHURST REGIONAL
6	ABERDARE	CESSNOCK
7	ABERDEEN	MUSWELLBROOK
8	ABERFOYLE	ARMIDALE REGIONAL
9	ABERGLASSLYN	MAITLAND
10	ABERMAIN	CESSNOCK
11	ABERNETHY	CESSNOCK
12	ABINGTON	URALLA
13	ACACIA CREEK	TENTERFIELD
14	ACACIA GARDENS	BLACKTOWN
15	ADAMINABY	SNOWY MONARO REGIONAL

Figure 8: Post Silver Layer Cleaned Data: Command: ***select*** *** ***from*** *dbt_mmenon_silver.lga_suburb_mapping lsm*

4.3 Gold Layer: Dimension and Fact Tables

In the Gold layer, we created dimension and fact tables in PostgreSQL using dbt to produce a curated, analytical-ready dataset. The Gold layer focuses on structuring data for reporting and analysis while linking it to the historical snapshot information.

Dimension Tables

1. **dim_host:** Contains host attributes such as host_id, host_name, host_since, host_is_superhost, and host_neighbourhood. It references the Silver layer snapshots to link each listing to the correct host state.
2. **dim_property:** Tracks property characteristics (property_type, room_type, accommodates) and uses current snapshot data to provide consistent keys for fact table joins.
3. **dim_roomtype:** Maintains room type information, linked to fact listings via dimension IDs.
4. **dim_lga:** Static dimension containing LGA codes and names, used to enrich listings with geographic information.

Each dimension table was materialized as a persistent, queryable table in PostgreSQL via dbt ({{ config(materialized='table', schema='gold', alias='dim_…') }}), enabling direct verification and reporting.

Fact Table

1. The fact_listing table consolidates Airbnb listing metrics and attributes, joining listings to dimension tables.
2. Foreign keys link each listing to current dimension records (dim_host_id, dim_property_id, dim_roomtype_id, dim_neighbourhood_id), ensuring referential integrity.
3. Metrics such as price, availability, review counts, and ratings are included for analytics.

Datamarts :
The Gold layer builds on Silver tables and snapshots to create analytical datamarts for Airbnb listings. Dimension tables (dim_host, dim_property, dim_roomtype, dim_lga) are joined to fact_listing, linking listings to current dimension states while preserving history. Views like dm_host_neighbourhood, dm_listing_neighbourhood, and dm_property_type aggregate metrics such as active listings, revenue, superhost rate, review scores, and pricing by month, neighborhood, or property type are derived.

4.4 Snapshots:

Snapshots were created for key Airbnb dimensions hosts, properties, and room types, using dbt to capture historical changes (SCD Type 2). Each snapshot defines a unique key (host_id, property_id, room_type_id) and check columns, so updates to relevant attributes trigger new

versions. Pulling the last 12 months of raw data, snapshots standardize text, handle nulls, and generate consistent IDs (e.g., MD5 hashes). This preserves dimension history.

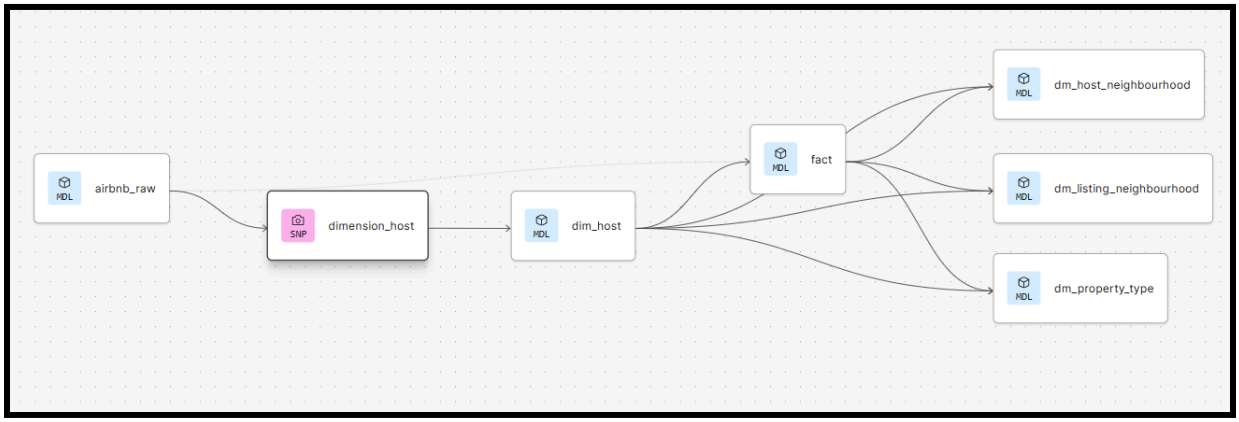


Figure 9: Snapshots for dimension_host

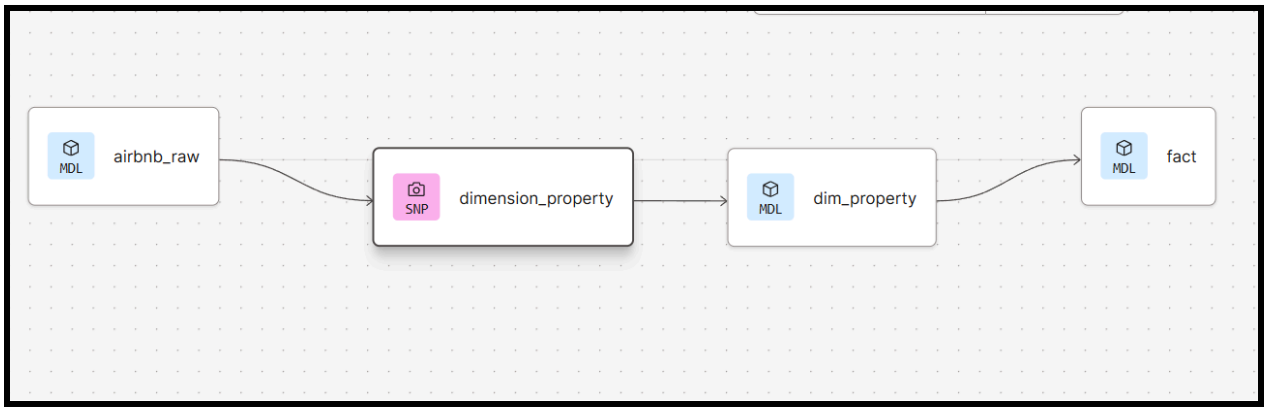


Figure 10: Snapshots for dimension_property

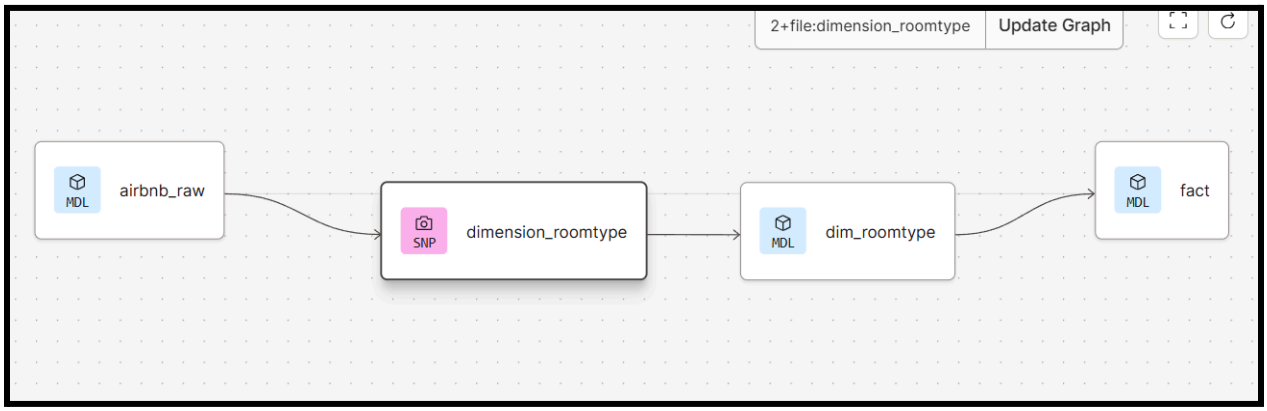


Figure 11: Snapshots for dimension_roomtype

Section 5. Dbt Job Creation for Data Ingestion:

1. A new production environment was set up for dbt by creating a dedicated workspace connected to the PostgreSQL warehouse and to load all the other month datas into the silver, gold pipelines.All changes to models, snapshots, and transformations were committed first to a feature branch and then merged into the main branch after review. This created new databases in Postgres, like postgres_bronze, postgres_silver, postgres_gold.
2. Jobs were deployed by manually running dbt build in the production environment, which executed the full sequence of model materializations, snapshot updates, and view creations, ensuring that the Silver and Gold layers were up-to-date and fully queryable.

Section 6. Part 4:Ad-hoc analysis

Question a) What are the demographic differences (e.g., age group distribution, household size) between the top 3 performing and lowest 3 performing LGAs based on estimated revenue per active listing over the last 12 months?

Insights: The top-performing LGAs tend to have smaller household sizes (2.4–2.7 vs. 2.9–3.2) and higher personal incomes, while the bottom performers have larger households and lower incomes. This suggests that Airbnb revenue is higher in more affluent areas with smaller households, likely reflecting greater demand for short-term rentals.

Recommendation: Focus marketing and property acquisition in higher-income, smaller-household LGAs, and consider tailored strategies (e.g., premium listings, targeted promotions) for underperforming areas to boost occupancy and revenue.

Bottom 3 Performers	Blacktown	52249	21536	180926	138302	3.2	672
Bottom 3 Performers	Burwood	7860	2484	11563	23421	2.9	552
Bottom 3 Performers	Strathfield	9302	2355	11783	25869	3	682
Top 3 Performers	Hunters Hill	1134	1425	10532	2723	2.7	977
Top 3 Performers	Mosman	3407	3000	22190	4306	2.4	1295
Top 3 Performers	Northern Beaches	29575	22440	201772	38204	2.7	916

Question b) Is there a correlation between the median age of a neighborhood (from Census data) and the revenue generated per active listing in that neighborhood?

Insights: The correlation of 0.62 indicates a moderately strong positive relationship between median age and revenue per active listing neighborhoods with older populations tend to generate higher Airbnb revenue. Areas with more mature residents may have higher disposable income or stable demand for short-term rentals.

Recommendation: Prioritize property listings and marketing efforts in neighborhoods with higher median ages to maximize revenue potential.

	123 correlation_coefficient
1	0.6221571109

Question c) What will be the best type of listing (property type, room type and accommodates for) for the top 5 "listing_neighbourhood" (in terms of estimated revenue per active listing) to have the highest number of stays?

Insights: Across the top 5 neighborhoods by estimated revenue per active listing, entire apartments for entire home/apt are consistently the most popular, with 2–4 accommodates yielding the highest total stay nights. Smaller units (2–4 guests) dominate in most areas, suggesting demand for compact, private stays.

Recommendations: Focus on acquiring or marketing entire apartments that cater to 2–4 guests in these high-revenue neighborhoods. Optimizing availability and pricing for these unit types could maximize total stays and revenue.

listing_neighbourhood	property_type	room_type	accommodates	num_listings	total_stay_nights
waverley	entire apartment	entire home/apt	2	14,992	387,004
northern beaches	entire apartment	entire home/apt	4	10,508	250,244
woollahra	entire apartment	entire home/apt	2	4,276	108,396
mosman	entire apartment	entire home/apt	2	1,508	36,560
hunters hill	entire apartment	entire home/apt	4	116	3,208

Question d)For hosts with multiple listings, are their properties concentrated within the same LGA, or are they distributed across different LGAs?

Insights: *The vast majority of hosts with multiple listings (**94.3%**) concentrate their properties within a single LGA, while only a small fraction (**5.7%**) spread their listings across multiple LGAs. This indicates that multi-listing hosts typically focus on dominating a specific local market rather than diversifying geographically.*

Business Recommendations: *Strategies targeting high-performing hosts should consider LGA-specific initiatives, such as localized promotions or support programs. Market interventions or incentives could be tailored per LGA to maximize occupancy and revenue from concentrated multi-listing hosts.*

	distribution_type	num_hosts	percent_of_multi_listing_hosts
1	Concentrated in one LGA	29,242	94.33
2	Distributed across multiple LGAs	1,759	5.67

Question e)For hosts with a single Airbnb listing, does the estimated revenue over the last 12 months cover the annualized median mortgage repayment in the corresponding LGA? Which LGA has the highest percentage of hosts that can cover it?

	lga_name	total_hosts	hosts_covering_mortgage	percentage_covering
1	BAYSIDE	22	0	0
2	BURWOOD	5	0	0
3	CANADA BAY	6	0	0
4	CANTERBURY-BANKSTOWN	3	0	0
5	BATHURST REGIONAL	1	0	0

Insights and Recommendations did not make sense because the code was unable to generate relevant data.

Section 7. Challenges, Issues and Resolutions

1. Setting up Airflow and ensuring smooth data ingestion into the Bronze layer was challenging. Developing and debugging the DAGs required multiple iterations and test runs to guarantee reliability. During this process, I implemented data sanity checks, standardized column names to lowercase, and ensured proper data type handling, which allowed the raw datasets to load consistently and correctly into PostgreSQL via Airflow.
2. Setting up production environment, committing changes, and deploying jobs reliably, maintaining sequential processing.
3. Setting up and running DBT jobs was challenging. DBT build was running my tables very slowly due to I needed to redo most of my code and fix it. Also had to understand how the Git structure works.
4. Understanding snapshots, dimensions, and fact tables was a challenge because these concepts were unheard of. Had to do quite a lot of research to be able to link and code it.
5. Overall, understanding the entire infrastructure itself was difficult.
6. In the Ad-Hoc analysis, part 5 of the code did not yield the desired results. I tried debugging it multiple times but failed.

Section 8. Conclusion, Recommendations, and Learnings

Conclusion:
This project successfully built a production-grade ELT pipeline for Airbnb and Census data in Sydney, transforming raw datasets into clean, standardized, and analytics-ready tables. The Bronze-Silver-Gold architecture, combined with snapshots and dbt transformations, enabled historical tracking, reliable aggregation, and the creation of BI-ready datamarts. Analytical insights highlighted key patterns in neighborhood revenue, demographics, and listing types, informing data-driven business decisions. The integration of multiple datasets allowed for a nuanced understanding of the market and reinforced the importance of maintaining data quality and consistency across layers.

Key Learnings:
Implementing the full ELT stack highlighted the importance of understanding medallion architecture, snapshots, and dimension-fact relationships for accurate historical tracking. Learnt using Airflow DAGs and managing dbt jobs and other data processing tools. The process emphasized meticulous data cleaning, standardization, and handling of edge cases to ensure reliability. Finally, integrating Census data with Airbnb listings reinforced the value of combining multiple datasets for richer, actionable business insights.